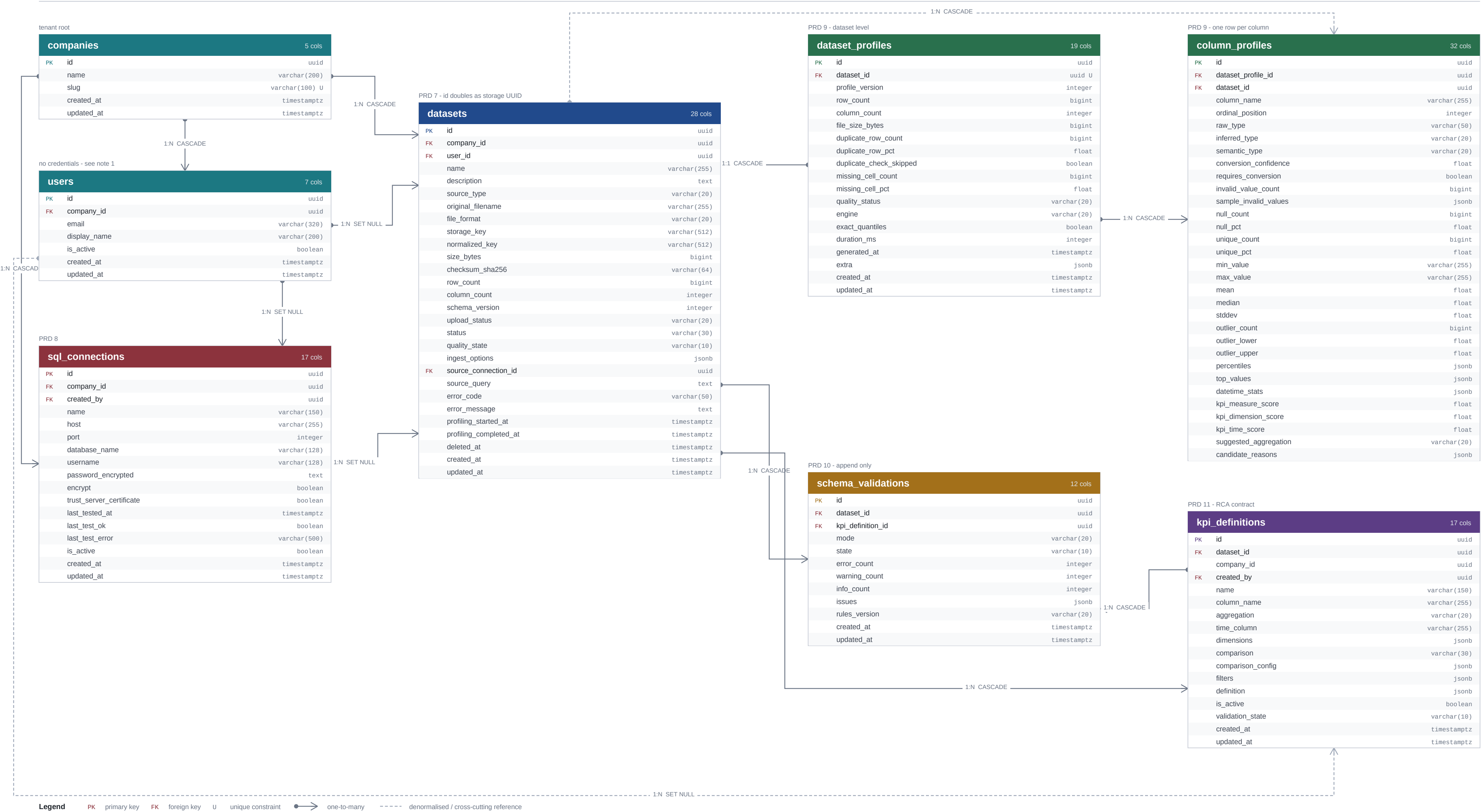


AI Root Cause Analysis Platform

Phase 1 - Data Foundation | entity relationship diagram

8 tables | 13 foreign keys

```
generated from backend/app/db/models/
```



Schema notes

Why the model is shaped the way it is

1 users carries no credentials.

Phase 1 ships without authentication by explicit decision, so there is no password, hash or session column anywhere in the schema. The table exists to satisfy the PRD's user_id metadata requirement and to give real auth somewhere to attach later.

2 datasets.id doubles as the physical storage UUID.

The storage key is {company_id}/{YYYY}/{MM}/{dataset_id}.{ext}. The original filename is kept in original_filename as metadata only and never contributes a character to a filesystem path (PRD principle 2).

3 Two different 'status' concepts, deliberately separated.

datasets.status is the pipeline state (pending_upload, uploaded, validating, profiling, profiled, analysis_ready, plus terminal upload_failed / profiling_failed / blocked). dataset_profiles.quality_status is the validation verdict (pass / warning / blocked). The PRD never enumerated either.

4 datasets.quality_state is a denormalised copy.

It mirrors the newest structural schema_validations.state so the dataset list can filter on quality without a join. schema_validations remains the source of truth.

5 column_profiles.dataset_id is denormalised too.

The authoritative parent is dataset_profile_id. Carrying dataset_id as well lets KPI-candidate queries rank columns without joining through dataset_profiles. Both are ON DELETE CASCADE, so the two paths cannot disagree.

6 Per-column statistics are rows, not a JSON blob.

One row per column lets the KPI candidate query rank in SQL and lets the Columns tab paginate server-side. Only the type-specific statistics - percentiles, top_values, datetime_stats, whose shapes differ entirely between numeric, categorical and datetime columns - stay in JSON.

7 kpi_definitions stores the contract twice, on purpose.

The typed columns drive validation and queries; definition holds the frozen, source-agnostic JSON contract handed to the RCA engine in exactly the shape PRD section 11 specifies. The service is the single writer and builds the JSON from the columns.

8 schema_validations is append-only.

Users re-upload corrected files and need to see what changed, so results accumulate rather than being overwritten. mode distinguishes a structural run from one re-checked against a specific KPI definition, which is why kpi_definition_id is nullable.

9 sql_connections has no plaintext password column.

password_encrypted holds a versioned Fernet token ('v1:' prefix) produced by app.core.security. The API read schema has no password field at all, so a credential cannot leak through a response even by mistake.

10 JSON columns are portable across dialects.

Every JSON column is declared JSON().with_variant(JSONB, 'postgres!'). Production keeps JSONB; the test suite runs the identical schema against SQLite.